



Global Supply Chain Performance Dashboard

Power BI Portfolio Project

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Power BI | SQL | Python | Excel | Tableau

1. Project Overview

The Global Supply Chain Performance Dashboard is an interactive Power BI solution designed to monitor and analyze key supply chain metrics across multiple regions. The dashboard provides insights into sales performance, profitability, order management, and delivery efficiency to support data-driven business decisions.

2. Business Problem

Organizations operating across multiple regions often struggle to track operational performance efficiently.

Key challenges include:

- Monitoring sales and profitability across regions
- Identifying high-performing and underperforming markets
- Tracking order volumes
- Measuring delivery performance
- Understanding business trends over time

This dashboard addresses these challenges by providing a centralized executive-level overview.

3. Project Objectives

Primary Objectives

- Analyze total sales performance
- Monitor profit and profit margins
- Track order volumes
- Evaluate on-time delivery performance
- Compare regional business performance
- Identify sales trends over time

4. Dataset Information

Dataset Type: Global Supply Chain Dataset

The dataset contains information related to:

- Orders
- Sales
- Products
- Customers
- Regions
- Profit
- Delivery Performance

Key Fields

- Order Date
- Category
- Order Region
- Sales
- Profit

- Order Quantity
- Delivery Status

5. Tools & Technologies Used

Power BI

- Dashboard Development
- Data Visualization

Power Query

- Data Cleaning
- Data Transformation

DAX

- KPI Calculations
- Custom Measures

Data Modeling

- Relationship Creation
- Star Schema Design

6. Data Preparation

The following steps were performed:

Data Cleaning

- Removed duplicate records
- Handled missing values
- Corrected data types
- Standardized region names

Data Transformation

- Created calculated columns
- Formatted date fields
- Optimized dataset structure

7. KPI Metrics

Total Sales

Measures total revenue generated.

Total Profit

Measures total profit earned.

Profit Margin %

Measures profitability.

Formula:

$$\text{Profit Margin \%} = \left(\frac{\text{Total Profit}}{\text{Total Sales}} \right) \times 100$$

Total Orders

Measures total order count.

Sales Growth

Measures change in sales over time.

On-Time Delivery %

Measures delivery efficiency.

8. Dashboard Features

Executive KPI Cards

- Total Sales
- Total Profit
- Profit Margin
- Total Orders
- Sales Growth
- On-Time Delivery %

Interactive Filters

- Year

- Category
- Region

Trend Analysis

Sales Trend chart displays monthly performance and seasonality patterns.

Regional Performance Analysis

Compare:

- Sales
- Profit
- Profit Margin

Across all operating regions.

9. Key Insights

Sales Performance

- Total Sales exceeded \$10.6M.
- Consistent sales performance observed throughout most months.

Profitability

- Total Profit reached \$1.3M.
- Average Profit Margin remained above 12%.

Regional Performance

- Western Europe and Central America generated the highest revenue.
- Certain regions showed higher profit margins despite lower sales volume.

Delivery Performance

- On-Time Delivery Rate stood at 45.55%.
- Indicates opportunities for logistics optimization.

10. Business Impact

This dashboard enables stakeholders to:

- Monitor business performance in real time

- Identify profitable regions
- Improve operational efficiency
- Optimize delivery processes
- Support strategic decision-making

11. Skills Demonstrated

✓ Power BI Dashboard Development

✓ Power Query

✓ DAX Measures

✓ Data Modeling

✓ KPI Design

✓ Business Intelligence

✓ Data Visualization

✓ Data Analysis

✓ Data Storytelling

Project Outcome

The project successfully transformed raw supply chain data into an interactive executive dashboard, providing valuable insights into sales, profitability, regional performance, and delivery efficiency.